

# Prajwal Kumar K

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[LinkedIn](#) | [Portfolio](#) | [GitHub](#) | [Codeforces](#) | [CodeChef](#) | [LeetCode](#) | [AtCoder](#)

## EDUCATION

**Indian Institute of Information Technology, Kottayam**

Kottayam, Kerala

*Bachelor of Technology in Computer Science and Engineering; CGPA: 8.66/10*

*Aug 2024 – May 2028 (expected)*

## TECHNICAL SKILLS

**Programming:** C++, Rust, Python, TypeScript, JavaScript, C, Java, SQL, Solidity, MATLAB

**AI/ML:** PyTorch, TensorFlow, scikit-learn, pandas, NumPy, Matplotlib

**Research and engineering:** Jupyter, Simulink, FastAPI, Next.js

**Research interests:** Algorithms and systems, game theory, machine learning and NLP, AI safety

**Languages:** English (Fluent), German (Business Proficiency), Hindi, Kannada

## EXPERIENCE

**Coders Club Coordinator**

Jan 2025 – Present

*IIIT Kottayam*

*Kottayam, Kerala*

- Organize programming contests with 80+ participants; curate problem sets and configure Codeforces Gym rounds.

**Openverse Technical Lead (Open Source Club)**

May 2025 – Jul 2026

*IIIT Kottayam*

*Kottayam, Kerala*

- Led technical work and PR reviews for 50+ Openverse contributors; authored Algorithmica sorting visualizations.

**Full-Stack Web Developer Intern**

Dec 2025 – Mar 2026

*CIO Mogul (Magazine)*

*Remote*

- Delivered 3 magazine sites in a 2-person team—[CIO Mogul](#), [CIO Mogul India](#), and [CIO Healthcare](#)—for publishing articles and digital issues.
- Automated PDF cover extraction, page navigation, and A3-spread splitting across 3 publications using Next.js/React and Supabase/PostgreSQL editorial workflows.

## PROJECTS

**PokeForge** | *Python, PyTorch, Gen 9 OU*

[Code](#)

- Prototyped a Gen 9 OU Pokémon agent evaluation testbed with 2 pinned baselines (in progress).
- Built PyTorch MLP/Transformer policies with legal-action masking; integrated FoulPlay/Metamon baselines and 61 offline contract tests, with simulator conformance still under evaluation.

**Bluff** | *Python, PyTorch, Bayesian modeling*

[Code](#)

- Designed a 2-player imperfect-information card game to study bluff detection and opponent adaptation.
- Implemented Bayesian opponent profiles and a learned public-information detector; improved draw-aware score by 4.9–5.8 percentage points across 8 synthetic profiles versus detector ablation.

**Drishti-XAI** | *Retinal imaging, MATLAB, Python*

- Evaluated an explainable retinal-screening prototype for generalization across 2 diabetic-retinopathy image datasets.
- Built the MATLAB/Python prototype and evaluation pipeline; at a fixed model and threshold, sensitivity/specificity were 0.908/0.877 on APTOS (n=733) and 0.363/0.865 on Messidor-2 (AUC 0.666).

**SiliconSpire** | *C++, Rust, Metaheuristics*

[C++](#) | [Rust](#)

- Engineered a Quadratic Assignment Problem solver to minimize weighted flow-distance cost for facility placement.
- Combined Grey Wolf Optimization and Tabu Search; reached published optima in 77/90 Rust runs (85.6%) versus 34/90 C++ runs across 3 small QAPLIB instances.

**BattleCP** | *Rust, Axum, Tokio, WebSockets*

[Live](#) | [Code](#)

- Developed a multiplayer Battleship game that awards turns for solving Codeforces problems.
- Implemented Rust/Axum WebSockets, Codeforces validation, server-side rules, and reconnects; recorded 6,922 cumulative page views.

## ACHIEVEMENTS AND CERTIFICATIONS

**Publication:** First-listed author and presenter, “Collaboration-Enabled Energy-Aware Learning Model Exchange (CE-LME) Framework for Edge Computing-Assisted Applications,” IEEE ICICI 2025. [Paper](#).

**Algorithmica:** selected for and presented at OOSC 3.0, IIT Kanpur (2025); team presentation at T4E 2025, IIT Madras.

**Hackathon:** Won Best Software Award at IEEE MACE .hack(); 2026 with Team Hogorithm (Anchor).

**Competitive programming:** Codeforces Pupil (peak 1221), CodeChef peak 1463, 400+ problems solved across platforms.

**Cyfrin Updraft (2025):** Blockchain Basics, Rust Programming Basics, Solidity Smart Contract Development, and Fundamentals of Zero-Knowledge Proofs.